3 Single Crystal Silicon and Wafer Preparation

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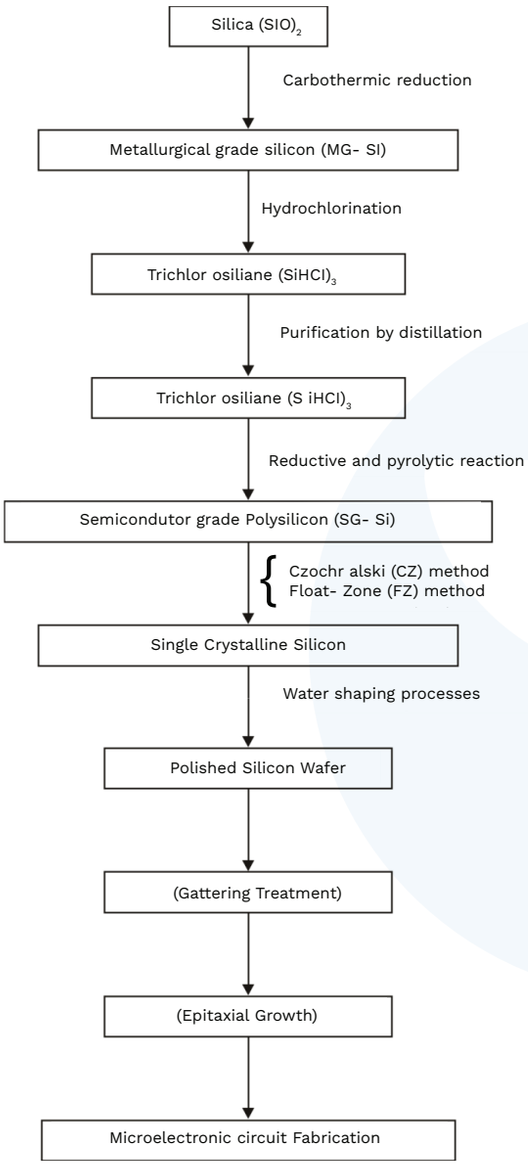


Fig. 3.1 Flow Diagram for a Typical Semiconductor Silicon Preparation Processes

- Silicon is among the most abundant elements on Earth. In total, 90 per cent of the Earth's crust is composed of silica.
- Following are the reactions that take place in the semiconductor grade silicon (SGS) production process.

$$SiO_2 + 2C \rightarrow Si + 2CO \rightarrow \text{Carbothermic reaction}$$

$$Si + 3HCl \rightarrow SiHCl_3 + H_2 \rightarrow \text{Hydrochlorination}$$

$$SiHCl_3 + H_2 \rightarrow Si + 3HCl \rightarrow \text{Pyrolysis}$$

- Float zone and Czochralski method are used to obtain single crystal silicon.
- Ingot trimming, orientation flattening, ingot etching, slicing, edge rounding, lapping, slice etching, and polishing are the wafer-shaping processes. After these processes, the wafers are inspected, and after the final inspection, they are packed.

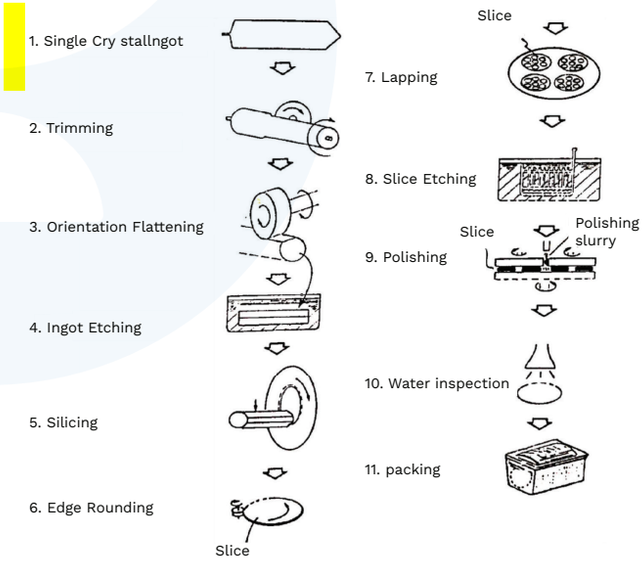


Fig. 3.2 Generic Wafer Shaping Processes

Float Zone Method

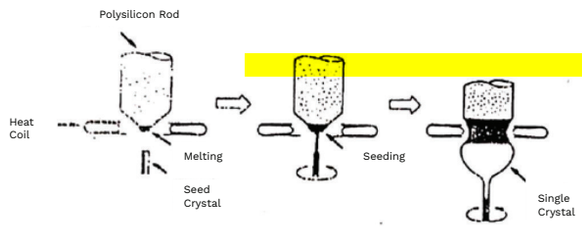


Fig. 3.3 Principle of Single Crystal Growth by Float Zone Method

- Molten zone is passed through a polysilicon rod to convert it into a single crystal ingot.
- In total, 20 per cent of single crystal silicon is produced by this method (Fz).
- A 150 mm diameter silicon crystal is obtained by this method.
- The tip of the polysilicon rod is contacted and fused with the seed with the desired crystal orientation. This process is called seeding.
- After seeding, a thin neck about 2–3 mm in diameter and 10–20 mm in length is formed. This process is called necking.

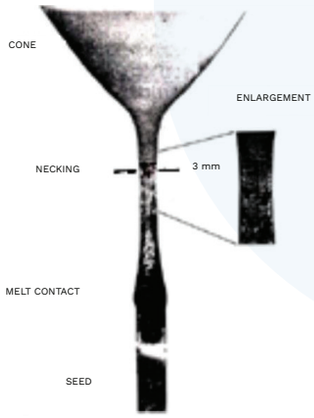


Fig. 3.4 X-Ray Topography of Seed, Neck, and Conical Part of Float Zone Silicon

- Dislocations introduced due to thermal shocks are eliminated by growing a neck. This necking process, called the dash technique, is fundamental for growing dislocation free crystals and is universally used for both FZ and CZ methods.
- FZ method produces single crystal silicon of high purity, high resistivity of 10–200 ohm cm, and oxygen impurity of 10^{16} atoms/cm^3.

Czochralski Method (Cz)

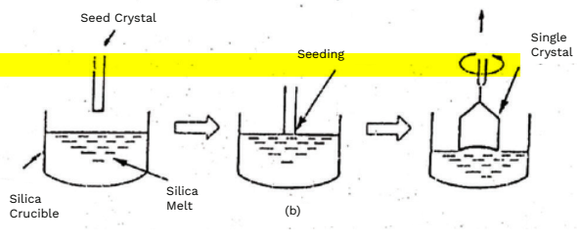


Fig. 3.5 Principles of Single Crystal Growth by Czochralski Method

- It is also called teal little method.
- In this method, a single crystal is produced by pulling from the silicon melt contained in a quartz crucible.
- In total, 80 per cent of single crystal silicon is produced by CZ.
- A 250–300 mm diameter silicon crystal is obtained by this method.
- Cz contains oxygen impurity of about 10^{18} atoms/cm_3.
- The resistivity of silicon 25 ohm cm or less.
- Regarding mechanical stress, it was found that FZ silicon is more vulnerable to thermal stress in the device fabrication process. Thus, CZ silicon is stronger than FZ silicon against thermal stresses. This is why, CZ silicon is used more in the fabrication of ICs.

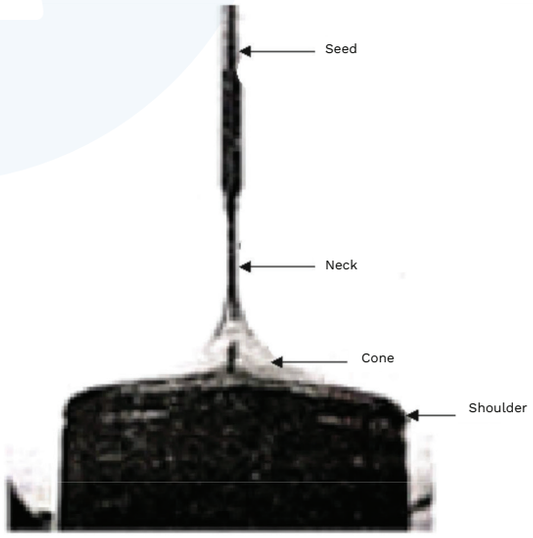


Fig. 3.6 Seed-End Part of As-Grown Czochralski Silicon Crystal

Practice Exercise

Level I

1. Which is the commonly used method for growing single crystal silicon?
(A) FZ method
(B) Stockbarger method
(C) CZ method
(D) Zone melting method

2. Which of the following reactions shows pyrolysis?
(A) $SiO_2 + 2C \rightarrow Si + 2CO$
(B) $Si + 3HCL \rightarrow SiHCL_3 + H_2$
(C) $SiHCL_3 + H_2 \rightarrow Si + 3HCL$
(D) $Si + O_2 \rightarrow SiO_2$

3. CZ refers to:
(A) cold zone method
(B) cathode zone method
(C) chemical zone method
(D) czochralski method

4. FZ stands for:
(A) Flowing Zone
(B) Float Zone
(C) Free Zone
(D) Fast Zone

5. $SiO_2 + 2C \rightarrow Si + 2CO$ is:
(A) hydro chlorination
(B) carbothermic reaction
(C) pyrolysis
(D) distillation

6. Teal little method is:
(A) CZ method
(B) FZ method
(C) zone melting method
(D) Stockbarger method

Level II

1. Which statement is correct for CZ method?
(A) Melt and growing crystals are usually not related.
(B) Melt and growing crystals are usually rotated counter clockwise.
(C) Melt and growing crystals are usually not rotated.
(D) Melt and growing crystals are usually kept at a constant position.

2. Necking process is also called:
(A) dash technique
(B) seeding
(C) thinning process
(D) orientation process

3. Dislocations introduced due to thermal shocks are eliminated by:
(A) growing a neck
(B) seeding
(C) chlorination
(D) pyrolysis

4. Oxygen impurity in CZ grown crystal silicon is:
(A) 10^{18} atoms/cm^3
(B) 10^{28} atoms/cm^3
(C) 10^{16} atoms/cm^3
(D) 10^{10} atoms/cm^3

5. A particular pulling rate is maintained in:
(A) FZ method
(B) CZ method
(C) zone melting method
(D) Bridgman method

6. What will be the next step after slicing (process) silicon wafers?
(A) lapping
(B) polishing
(C) cleaning
(D) grinding

Answer Key

Level I

1. (C) **2.** (C) **3.** (D) **4.** (B) **5.** (B) **6.** (A)

Level II

1. (B) **2.** (A) **3.** (A) **4.** (A) **5.** (B) **6.** (A)